

Lesson 26 — How Machines Learn from Data

Matches LEGO Teacher Portal lesson: "Dance Partner".

Objective (I can...): I can explain how a classifier learns from examples, collect a small dataset, and run a first training. **CS concept:** Supervised learning basics; a dataset as labeled examples · **Builds on:** Lesson 25 (we'll wrap AI calls in functions and conditions). *Privacy: training runs in the browser on-device via Google Teachable Machine — no login, and images aren't stored unless someone saves them.*

AI Connection (2 min): This is the real thing: you supply labeled examples, and the model learns the pattern that separates them — the honest version of every "AI learns from examples" note this year. *First-principles question:* "How would you prove to a skeptic that the model actually learned, and didn't just memorize your face?"

Materials: Device per pair with camera and browser (Google **Teachable Machine**), LEGO CS & AI kit, notebooks set up as a **dataset log** (category → number of examples → notes). **This is the AI unit's build lesson** — build the AI responder model now and keep it assembled through Lesson 29.

60-minute flow - Warm-up (8): "Be the classifier" — coach shows mixed picture cards; pairs sort into two categories and write the rule they used. Point: a classifier learns a rule like that, but from many examples, not a sentence. - **Concept + demo (10):** Walk through the honest pipeline on Teachable Machine: create two classes, add examples, train, test. Say clearly what it is (a simple image classifier) and what it is NOT (not building a chatbot or "its own AI"). Restate the privacy facts aloud. - **Build/Code (30):** Pairs **build the AI responder model** (the unit's one build), then create a 2-category image classifier in Teachable Machine, capture several examples each, train, and test it live. They start the **dataset log** in the notebook (examples per category). *Swap at 15 min so both capture and operate.* - **Challenge (7):** Add a third category; note whether accuracy on the first two changed. - **Share + pack-up (5):** Pairs show their classifier labeling live and read their dataset log. **Leave the model built.** Close the Teachable Machine tab (nothing saved).

Pair roles: Driver captures and labels examples, Navigator runs tests and keeps the dataset log. Swap at 15 min. *(Trio rotates capture/train/test.)*

Differentiation: *Support* — two very distinct categories (bright card vs. dark card). *Extension / fast-finisher* — add examples from varied angles and lighting. *Reteach* — capture examples together and watch the confidence bars rise as examples are added.

Exit check: Pair trains a working 2-category classifier and can explain, honestly, what it learned and what it is not. **At-home (optional):** List three everyday AI classifiers (spam filter, photo tagging, voice wake-word) and what examples each likely learned from. ~10 min.